Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Independent

1. Time and rate

Explain why 90 s gives a larger relative rate than 180 s under matched conditions.

Relative rate = 1 ÷ time, for the same starting amount and endpoint.

2. Read the endpoint

At 60 s starch is detected; at 90 s it is not. What endpoint value is used in the rate calculation?

Use the first observed endpoint, not an invented time between samples.

3. Show the working

Calculate relative rate for an endpoint of 120 s; show the equation, substitution and unit.

Relative rate = 1 ÷ time. A value to three significant figures is sufficient here.

4. Match the instruction

Distinguish DESCRIBE from EXPLAIN when answering about a graph.

Describe = state the pattern. Explain = link that pattern to scientific reasons.

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Exit Ticket · Independent

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Work out both relative rates from the first series. Use them to compare the results. Set a limit on what this shows.

Show 1 ÷ time and s^{-1}. End with “in this series”.

2. Use the repeat or the sampling interval. Give one limit of the test. Then give one change for the next test. Say why it would help.

Name the limitation before an improvement. A shorter interval must be authorised and feasible.

Under the heading Exit ticket: answer the short questions in your book.